

Zhongtao (Tony) Guan

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RESEARCH INTERESTS

Autonomous Systems Co-design, Aerial Robotics, Microrobotics, Hybrid Locomotion.

EDUCATION

Massachusetts Institute of Technology

2025 – 2030 (expected)

Ph.D. Student in Electrical Engineering & Computer Science

Cambridge, MA, U.S.

ShanghaiTech University

2021 – 2025

B.Eng. in Electronic Information Engineering (President's List)

Shanghai, China

RESEARCH EXPERIENCE

Swarm Flight at the Insect-Scale

Sep. 2025 – Present

Soft and Micro Robotics Laboratory, MIT

Advisor: Prof. Kevin Chen

- Upgraded the Vicon motion capture system and communication stack to provide reliable, low-latency state feedback alongside 20 high-rate amplifier drive signals without data loss.
- Used MathWorks tools to rapidly prototype and validate control, signal-processing, and monitoring pipelines across the tightly coupled electromechanical hardware stack.
- Built infrastructure for waveform generation, actuator health monitoring, and precise state feedback to support upcoming swarm flight experiments with up to five insect-scale robots for cooperative payload transport and confined-space inspection.

Sensor Autonomy for Insect-Scale Flapping-Wing Robots [S.1]

Jul. 2024 – Dec. 2024

Soft and Micro Robotics Laboratory, MIT

Advisor: Prof. Kevin Chen

- Designed a sensory suite, including IMU, distance sensor, and optical flow camera for onboard state estimation.
- Developed microfabrication methods to integrate miniature sensors on a 1 cm × 1 cm board weighing under 0.24 grams.
- Achieved 10-second hovering flight of an insect-scale aerial robot using altitude and attitude feedback.

Sub-Gram Hopper with Passive Leg and Flapping-Wing [J.1]

Feb. 2024 – Sep. 2024

Soft and Micro Robotics Laboratory, MIT

Advisor: Prof. Kevin Chen

- Showcased the robot's capabilities in overcoming obstacles, navigating various terrains, and performing somersaults.
- Constructed high-fidelity simulation environments in MuJoCo and Simscape to mimic fast dynamics of the robot.
- Performed trajectory optimization and nonlinear model predictive control in contact-rich environments.

A 2-DoF Backdrivable Leg Prototype Design [C.1]

Aug. 2023 – Jan. 2024

Motor and Motion-Control Lab, ShanghaiTech

Advisor: Prof. Jiahao Chen

- Designed linear actuators integrating AC motors, 12-phase motor drives, and ball screws for bipedal robots.
- Developed robot joints using nonlinear four-bar linkages, and optimized torque output from human gait data.
- Showcased the robot's backdrivability and minimal clearance for energy recovery and simplified control without joint encoders.

Single-ended Fault Location for Hybrid Transmission Lines [C.2]

Jun. 2022 – Jan. 2023

Power System Protection and Automation Laboratory, ShanghaiTech

Advisor: Prof. Yu Liu

- Refined the classical Eriksson method for analytical fault location results on hybrid transmission lines.
- Utilized fully distributed line models and pseudo-solution identification strategies for better accuracy.
- Demonstrated fault location error < 1% for both high and low fault resistance cases.

PUBLICATIONS & MANUSCRIPTS

J: journal article; C: conference paper; S: manuscript. +Equal contribution.

- [S.1] Yi-Hsuan Hsiao⁺, Quang Phuc Kieu⁺, **Zhongtao Guan⁺**, Suhan Kim, Jiaze Cai, Owen Matteson, Jonathan P. How, Elizabeth Farrell Helbling, and Yufeng Chen. **Controlled Flight of an Insect-Scale Flapping-Wing Robot via Integrated Onboard Sensing and Computation**. arXiv:2602.08328, 2026. Pre-accept at *Science Robotics*
- [J.1] Yi-Hsuan Hsiao⁺, Songnan Bai⁺, **Zhongtao Guan⁺**, Suhan Kim, Zhijian Ren, Pakpong Chirarattananon, Yufeng Chen. **Hybrid Locomotion at the Insect scale: Combined Flying and Jumping for Enhanced Efficiency and Versatility**. *Science Advances*, 2025
- [C.1] **Zhongtao Guan**, Yiming Chen, Junlei Zhu, Yu Hu, Weibang Bai, and Jiahao Chen. **Preliminary Result of Cury: A Backdrivable Leg Design Using Linear Actuators**. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
- [C.2] **Zhongtao Guan**, Jun Wan, Yu Liu, Dian Lu, Mengzhao Duan, Renke Huang. **Accurate Single-Ended Fault Location for Cable-OHL Hybrid Transmission Lines**. *IEEE Power and Energy Society General Meeting (PESGM)*, 2023.
- [C.3] Jiayu Yang, Yu Liu, Kang Yue, **Zhongtao Guan**, et al. **Closed-Form Solutions of Mutual Inductance and Load for LCC-S Wireless Power Transfer Systems**. In *3rd IEEE International Conference on Industrial Electronics for Sustainable Energy Systems*, 2023.
- [C.4] Mengzhao Duan, Yu Liu, Ze Liu, Xinchun Zou and **Zhongtao Guan**. **A Group of Single-Ended Time-Domain Line Fault Location Methods Using Breaker Operation Information**. In *IEEE Power and Energy Society General Meeting (PESGM)*, 2023.

HONORS & FELLOWSHIPS

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| • MathWorks Fellowship, Massachusetts Institute of Technology | 2026–2027 |
| • Grass Instrument Company Fellowship, Massachusetts Institute of Technology | 2025 |
| • Undergraduate National Exchange Fellowship, ShanghaiTech University | 2024 |
| • International Conference Fellowship, ShanghaiTech University | 2023 |
| • Outstanding Teaching Assistant, ShanghaiTech University | 2023 |

PRESENTATIONS & OUTREACH

- **Microsystems Annual Research Conference:** "Controlled Flight of an Insect-Scale Flapping-Wing Robot via Integrated Onboard Sensing and Computation" 2026
- **Microsystems Annual Research Conference:** "Hybrid Locomotion at the Insect-Scale: Combined Flying and Jumping for Enhanced Efficiency and Efficacy" 2025
- **ShanghaiTech Automation and Robotics Center Annual Seminar:** "A Backdrivable Leg Design Using Linear Actuators" 2025

SELECTED PRESS

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| • IEEE Spectrum: Tiny robot bug hops and jumps | 2025 |
| • Tech Briefs: Hopping robots to traverse challenging terrain | 2025 |
| • MIT News: Hopping gives this tiny robot a leg up | 2025 |
| • Veritasium: Why is MIT making robot insects? | 2025 |

SKILLS

- **Mechatronic Design:** KiCad, Altium Designer, SolidWorks
- **Simulation Tools:** MuJoCo, Simscape, Drake, Webots, Simulink Real-Time

REFERENCES

- **Research Supervisor:** [Kevin Chen](#), Associate Professor, MIT, Contact: yufengc@mit.edu
- **Research Supervisor:** [Jeffrey H. Lang](#), Professor, MIT, Contact: lang@mit.edu
- **Research Supervisor:** [Yu Liu](#), Associate Professor, ShanghaiTech, Contact: liuyu@shanghaitech.edu.cn
- **Research Supervisor:** [Jiahao Chen](#), Assistant Professor, ShanghaiTech, Contact: chenjh2@shanghaitech.edu.cn